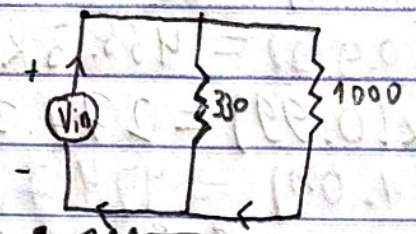
Pedro Umcovsk:

EDES 301 Assignment #1

1)

a) Red, red, black, gold
b) Orange, orange, brown, gold
c) Brown, black, red, gold
d) Red, red, red, gold
e) Yellow, violet, red, gold
f) Brown, black, orange, gold
g) Brown, black, yellow, gold

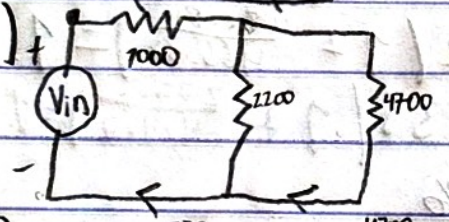
2) a)



$$\frac{1}{330} + \frac{1}{1000} = \frac{1}{R_{eq}}$$

$$R_{eq} \cong 250\ \Omega$$

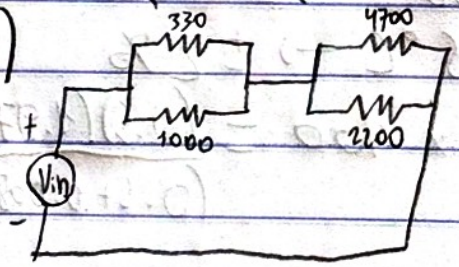
b)



$R_1 = 1000,\ R_2 = 2200,\ R_3 = 4700$

$$R_{eq} = (R_1 + (R_2 \| R_3)) \cong 2498\ \Omega$$

c)



$$(330 \| 1000) + (4700 \| 2200) \cong 1746\ \Omega$$

3) $R_{1max} = 330(1.05) = 346.5\ \Omega$

$R_{1min} = 330(0.95) = 313.5\ \Omega$

$R_{2max} = 1000(1.05) = 1050\ \Omega$

$R_{2min} = 1000(0.95) = 950\ \Omega$

$R_{1min} \| R_{2min} = 235.71\ \Omega$

$R_{2max} \| R_{2max} = 260.53\ \Omega$

4) $R = \frac{V_s - V_{max}}{I_f} = \frac{3.3 - 2.0}{10mA} = 130\,\Omega$

5-a) $\frac{R_1}{R_2} = \frac{280}{470} = 0.5957$

$V_{out} = 1.24(1 + 0.5957) = 1.978V \cong 2.0V$

b) $R1_{max} = 280K(1.01) = 282.8k$

$R_{amin} = 470k(0.99) = 465.3K$

$R1_{min} = 280k(0.99) = 277.2K$

$R_{max} = 470k(1.01) = 474.7k$

$V_{out_{max}} = 1.242(1 + \frac{282.8k}{465.3k}) = 1.9856V$

$V_{in_{max}} = 1.235(1 + \frac{277.2}{474.7}) = 1.9562V$

c) $25°C \rightarrow \pm 1\%$

$-40°C$ to $125°C \rightarrow \pm 2\%$

d) $Eff. = \frac{I_o V_o}{(I_o + I_q)V_i} \times 100 = \frac{(0.5)(1.97)}{(0.5 + 0.008)5} = 38.8\%$